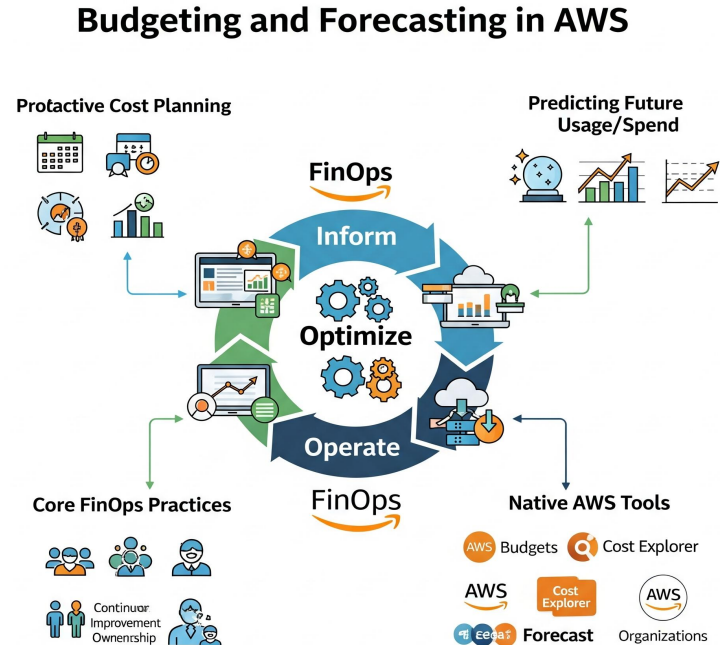


AWS FinOps

Budgeting and Forecasting

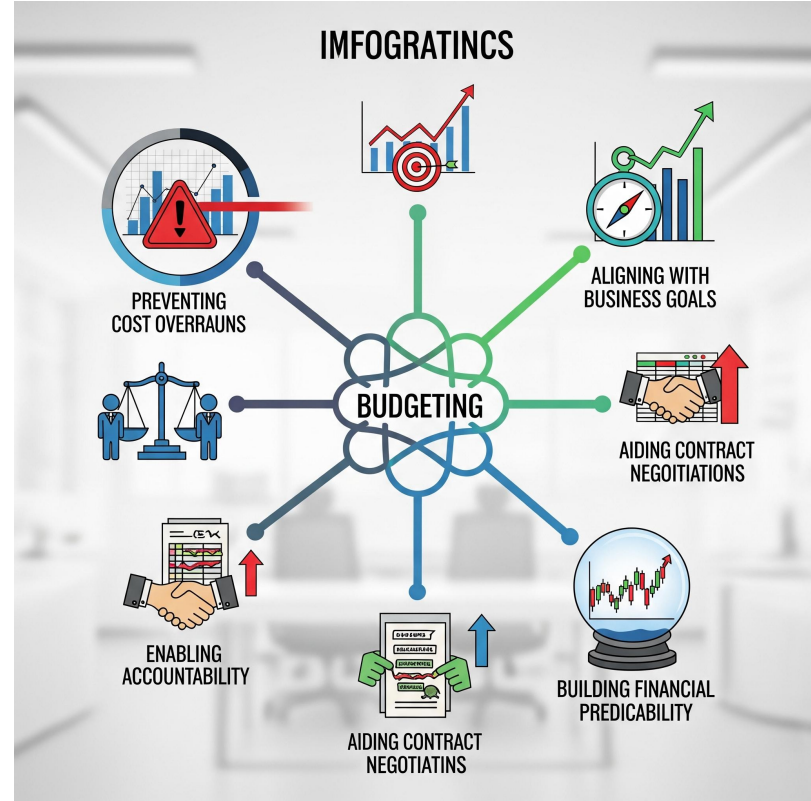
Budgeting and Forecasting in AWS FinOps

- ❖ Budgeting = proactive cost planning
- ❖ Forecasting = predicting future usage/spend
- ❖ Both are core FinOps practices
- ❖ AWS provides native tools for both
- ❖ Supports Inform, Optimize, Operate cycle



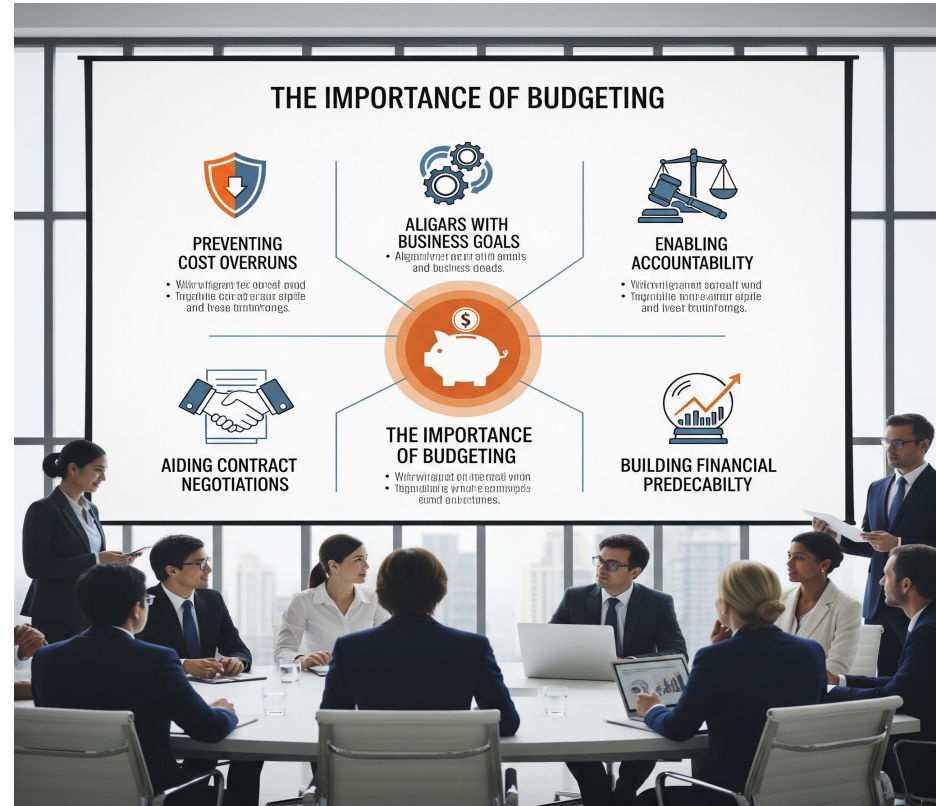
Importance of Budgeting

- ❖ Prevents cost overruns
- ❖ Ensures alignment with business goals
- ❖ Enables accountability across teams
- ❖ Helps in contract negotiations
- ❖ Builds financial predictability



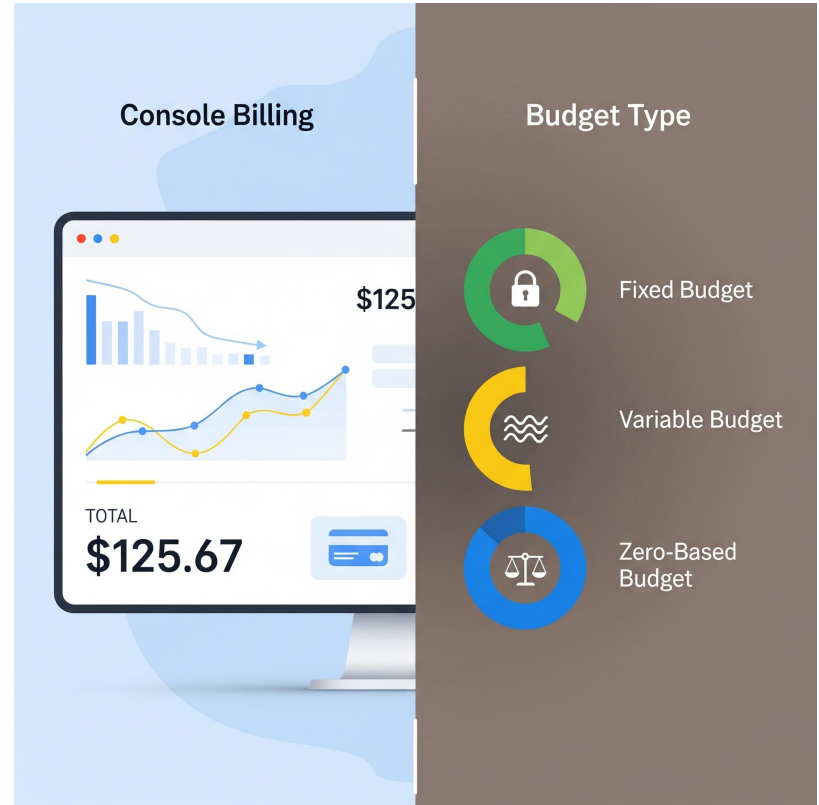
AWS Budgets Overview

- ❖ Define custom budgets (cost, usage, RI/SP coverage)
- ❖ Set alerts when thresholds are reached
- ❖ Automate actions (stop resources, notify)
- ❖ Integrates with AWS Organizations
- ❖ Supports forecasting



Creating AWS Budgets

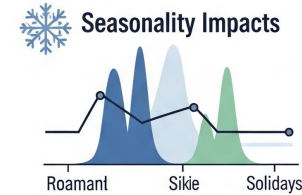
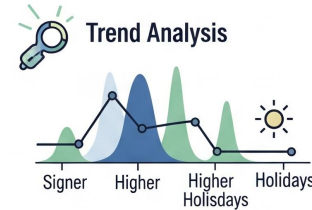
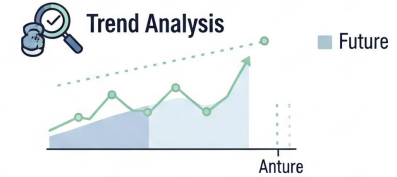
- ❖ Console → Billing → Budgets
- ❖ Select budget type:
Cost/Usage/RI
- ❖ Define threshold + email/SNS alerts
- ❖ Scope: linked accounts or tags
- ❖ Validate in Cost Explorer



Forecasting Basics

- ❖ Use historical billing data
- ❖ Apply trend analysis
- ❖ Seasonality impacts forecasts
- ❖ Tie forecasts to product roadmaps
- ❖ Update forecasts continuously

Forecasting Basics



AWS Cost Explorer Forecasting

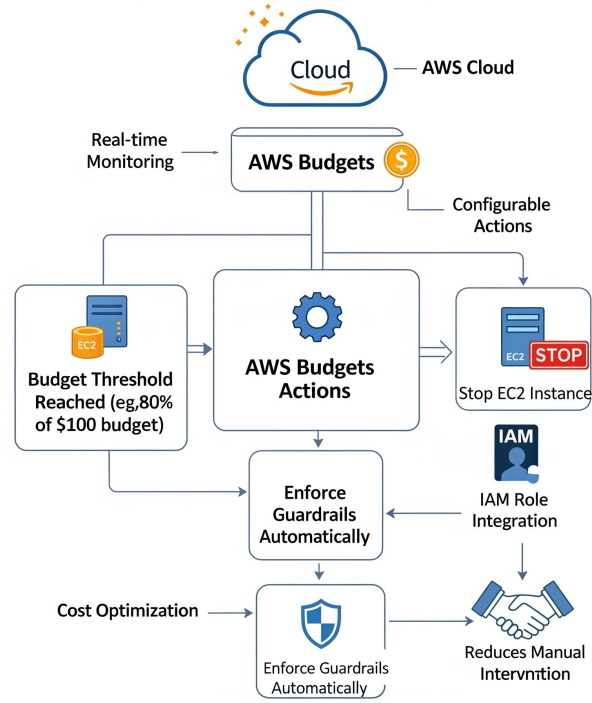
- ❖ Provides 3-month forecast by default
- ❖ Based on usage patterns
- ❖ Visual charts for decision-making
- ❖ Supports filtering by service/account/tag
- ❖ Export for Finance reviews



Budget Automation in AWS

- ❖ Use **Budgets Actions**
- ❖ Example: stop EC2 at threshold
- ❖ Supports IAM role integration
- ❖ Enforce guardrails automatically
- ❖ Reduces manual intervention

Budget Automation in AWS

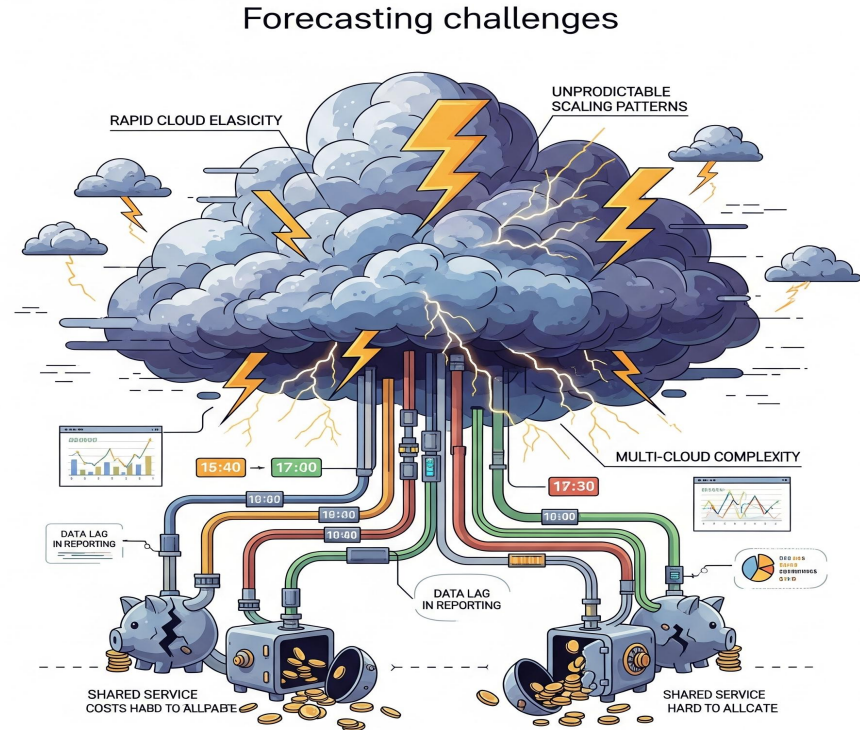


Forecasting Challenges

- ❖ Rapid cloud elasticity
- ❖ Unpredictable scaling patterns
- ❖ Shared service costs hard to allocate
- ❖ Data lag in reporting
- ❖ Multi-cloud complexity

Case Study – Retail Company

- ❖ Holiday traffic spike caused overspend
- ❖ Solution: budgets + Cost Explorer forecast
- ❖ Result: 20% cost savings
- ❖ Improved alignment with marketing campaigns
- ❖ No downtime



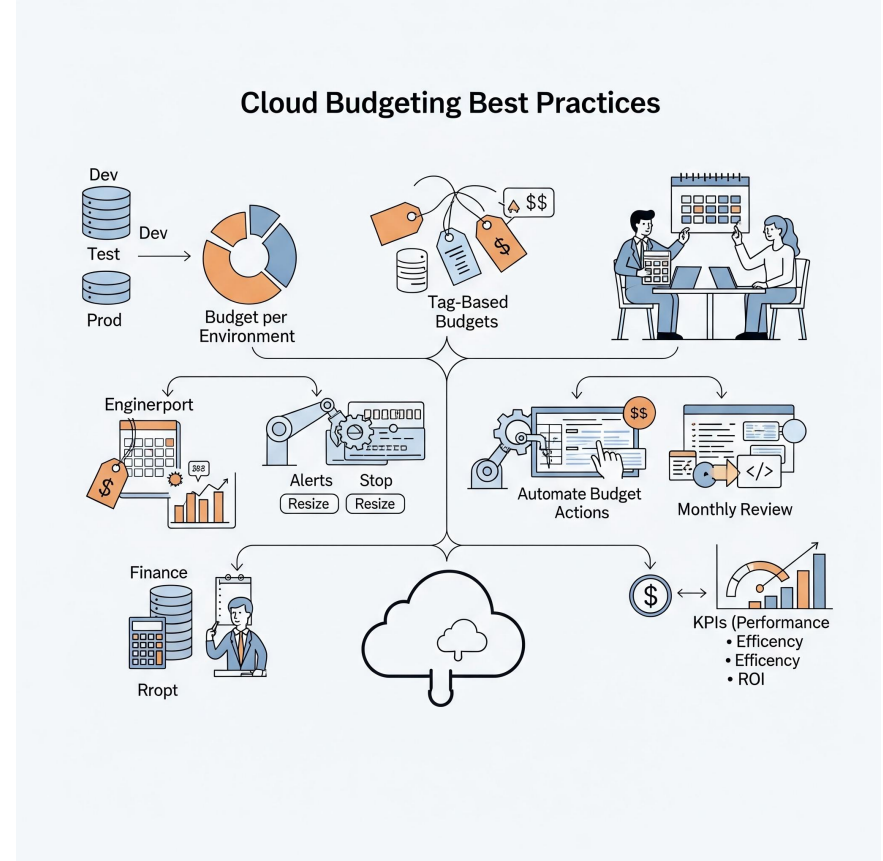
Case Study – SaaS Startup

- ❖ Problem: variable per-customer cost
- ❖ Forecast by customer segment
- ❖ Optimized infra for predictable costs
- ❖ Budget per product line
- ❖ Increased investor confidence



Best Practices – Budgeting

- ❖ Create budgets per environment (Dev/Prod)
- ❖ Apply tag-based budgets
- ❖ Automate via Budgets Actions
- ❖ Review monthly with Finance + Engg
- ❖ Tie budgets to KPIs



Summary-Budgeting and Forecasting

- ❖ AWS Budgets = guardrails
- ❖ Cost Explorer = forecasting tool
- ❖ Automations reduce risk
- ❖ Use cases: retail, SaaS, enterprise
- ❖ Supports proactive FinOps culture

KPIs & Metrics

Why Track KPI in FinOps

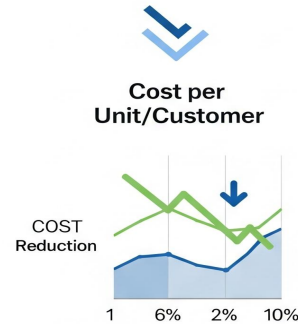
- ❖ Measure cost efficiency
- ❖ Drive accountability
- ❖ Benchmark improvements
- ❖ Align with business outcomes
- ❖ Support maturity journey



Common FinOps KPIs

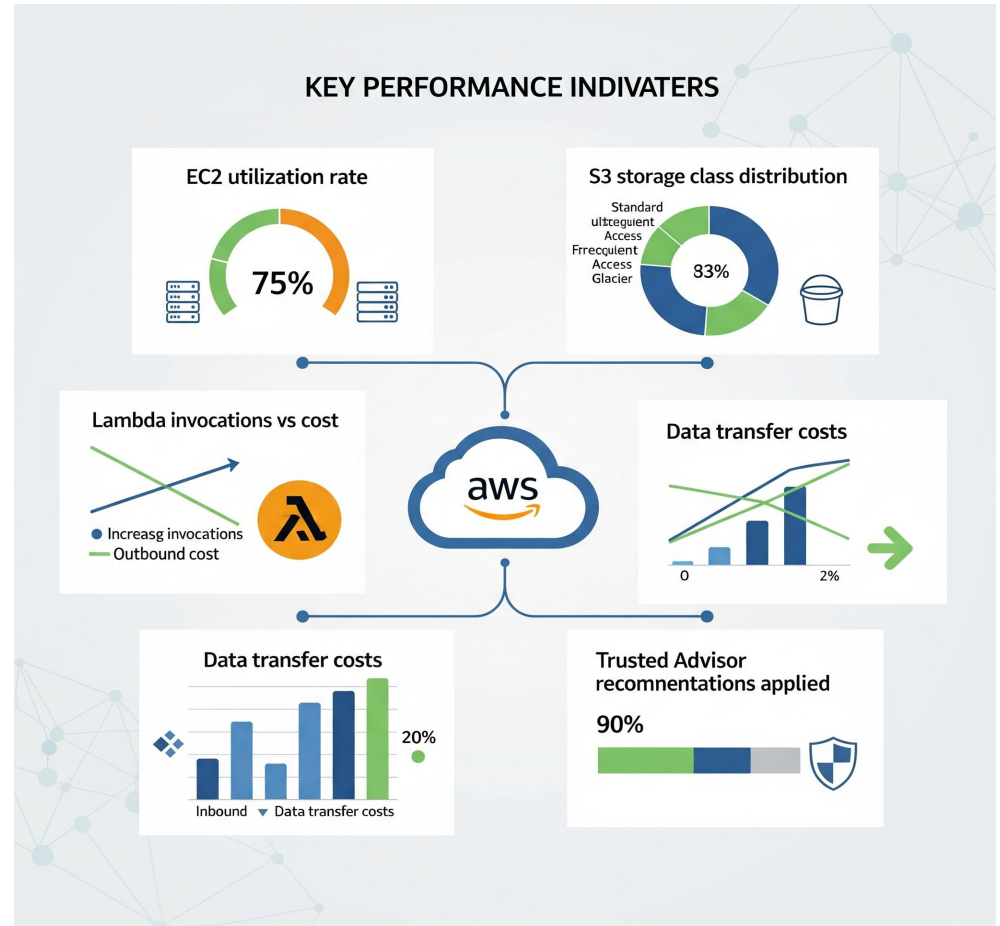
- ❖ Cost per unit/customer
- ❖ % Reserved Instance (RI) coverage
- ❖ % Savings Plan utilization
- ❖ Budget adherence %
- ❖ Tag compliance %

Key FinOps KPIs



AWS-Specific KPIs

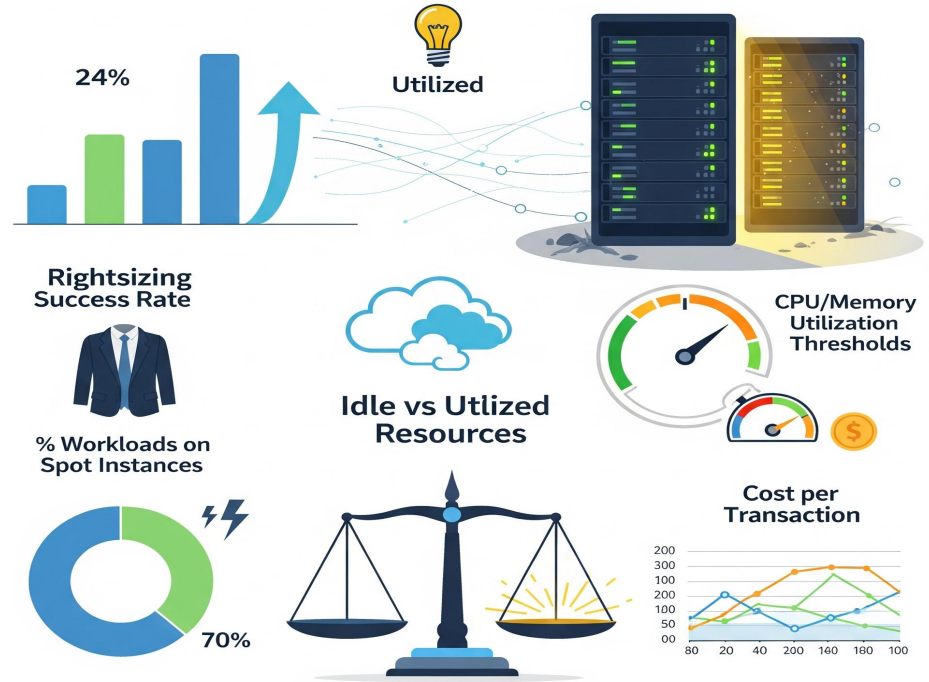
- ❖ EC2 utilization rate
- ❖ S3 storage class distribution
- ❖ Lambda invocations vs cost
- ❖ Data transfer costs
- ❖ Trusted Advisor recommendations applied



Efficiency Metrics

- ❖ Idle vs utilized resources
- ❖ Rightsizing success rate
- ❖ % workloads on spot instances
- ❖ CPU/memory utilization thresholds
- ❖ Cost per transaction

Cloud Cost Efficiency

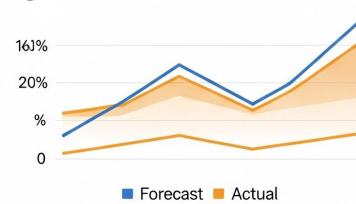


Forecast Accuracy Metrics

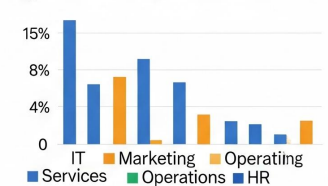
- ❖ Forecast vs actual spend variance
- ❖ Accuracy per service/team
- ❖ Rolling average deviations
- ❖ % budgets exceeded
- ❖ Predictability score

Forecast Accuracy Metrics

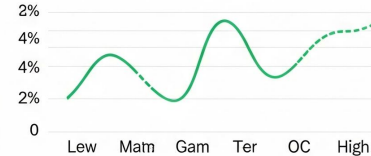
③ Forecast vs Actual Spend Variance



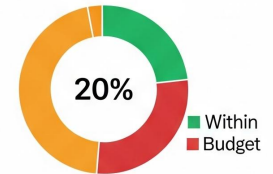
⚙️ Accuracy per Service/Team



⚙️ Rolling Average Deviations
Rolling Average Deviations



📁 % Budgets Exceeded



🕒 Predictability Score

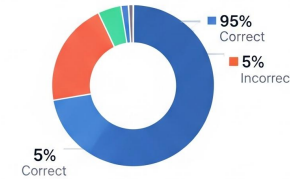


Governance Metrics

- ❖ % resources tagged correctly
- ❖ Number of policy violations
- ❖ Budget automation success rate
- ❖ Compliance with guardrails
- ❖ Policy-driven cost savings

Governance Metrics

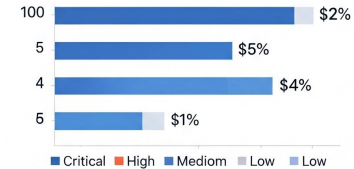
Correctly Tagged Resources



Budget Automation Success Rate



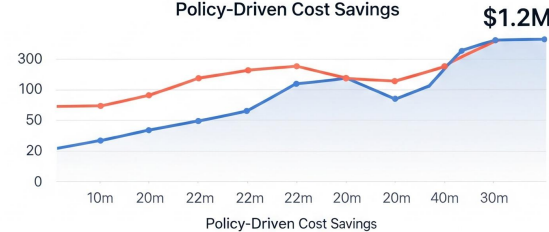
Policy Violations



Compliance with Guardrails

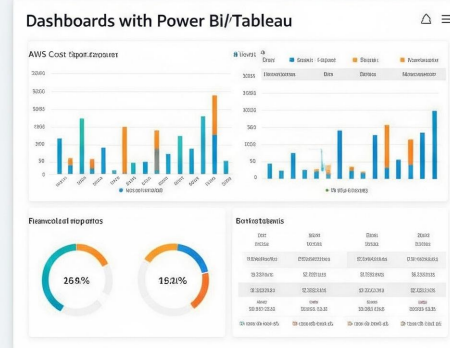
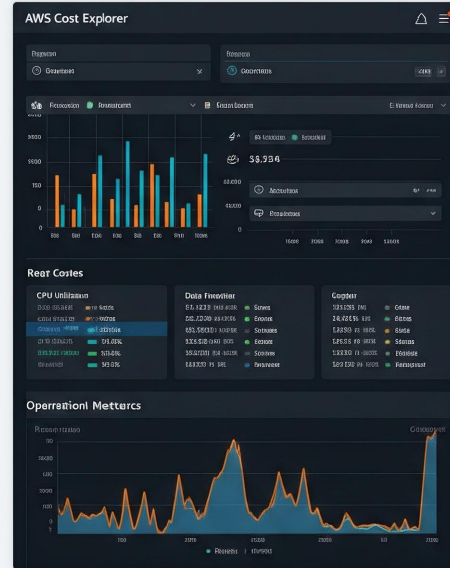


Policy-Driven Cost Savings



Dashboards for KPIs

- ❖ AWS Cost Explorer dashboards
- ❖ CloudWatch custom metrics
- ❖ Third-party FinOps platforms
- ❖ Power BI/Tableau integration
- ❖ Multi-account org dashboards



Case Study-SaaS KPI Tracking

- ❖ KPI: cost per active customer
- ❖ Optimized DB tiering reduced costs
- ❖ Dashboard showed real-time unit economics
- ❖ Product prioritized features with cost in mind
- ❖ Result: improved profitability



Best Practices-KPIs

- ❖ Define KPIs jointly across personas
- ❖ Automate data collection
- ❖ Use tags for granularity
- ❖ Align KPIs to business outcomes
- ❖ Iterate regularly



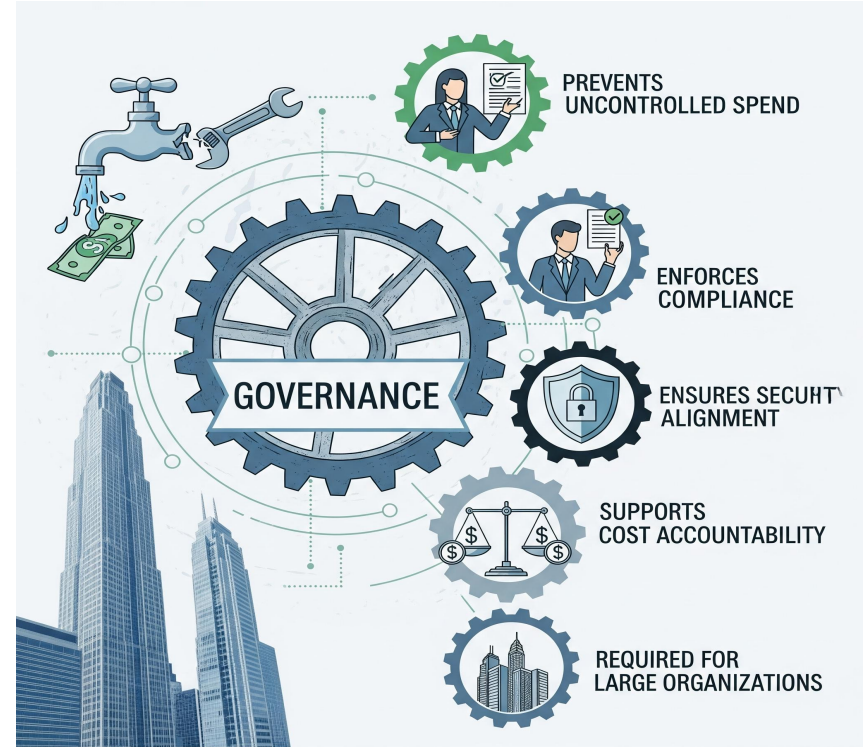
Summary-KPIs

- ❖ KPIs = accountability drivers
- ❖ AWS provides rich metrics
- ❖ Must align with unit economics
- ❖ Dashboards empower transparency
- ❖ Key for FinOps maturity

Governance and Policy Enforcement

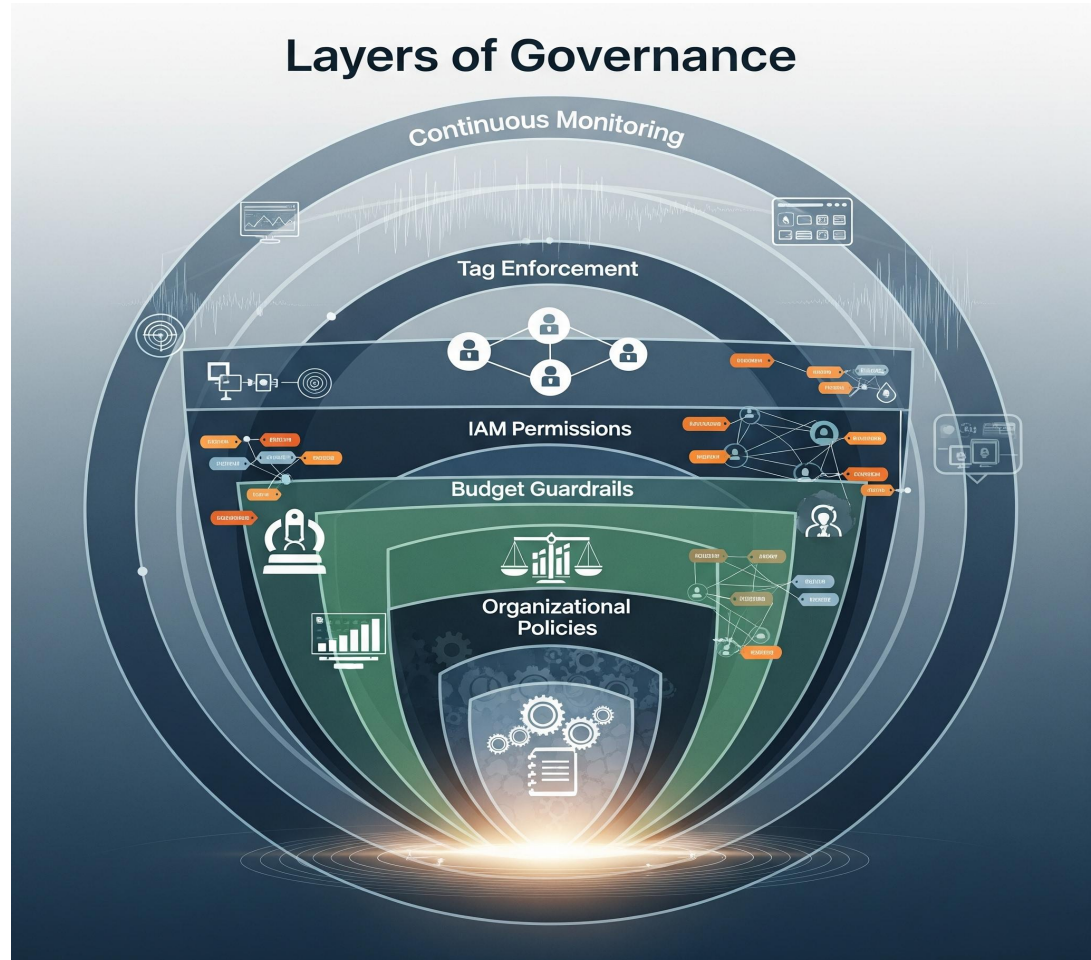
Importance of Governance

- ❖ Prevents uncontrolled spend
- ❖ Enforces compliance
- ❖ Ensures security alignment
- ❖ Supports cost accountability
- ❖ Required for large orgs



Layers of Governance

- ❖ Organizational policies
- ❖ Budget guardrails
- ❖ IAM permissions
- ❖ Tag enforcement
- ❖ Continuous monitoring



AWS Service Control Policies (SCPs)

- ❖ Applied at AWS Org level
- ❖ Restrict services/accounts
- ❖ Example: deny expensive instance types
- ❖ Supports governance at scale
- ❖ Non-bypassable



Service Control Policies (SCPs)

Effect: Deny

Action: **

[restricy Colicies **[**]]]

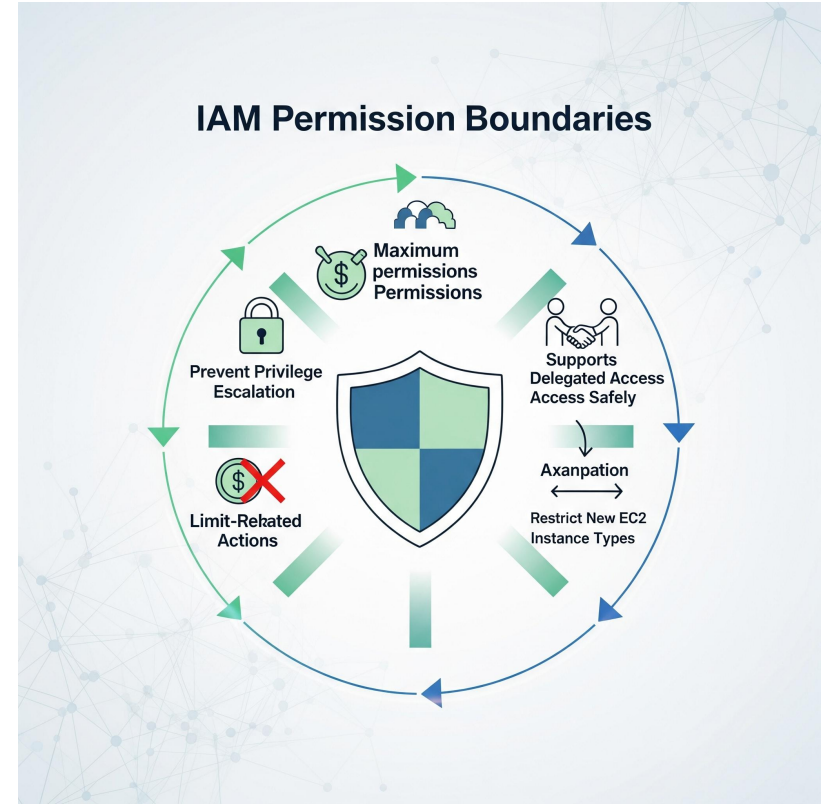
Resource: **

[resticcPolicies **[**]]]

Resource: **

IAM permission Boundaries

- ❖ Define max permissions for IAM roles
- ❖ Prevent privilege escalation
- ❖ Limit cost-related actions
- ❖ Supports delegated access safely
- ❖ Example: restrict new EC2 instance types



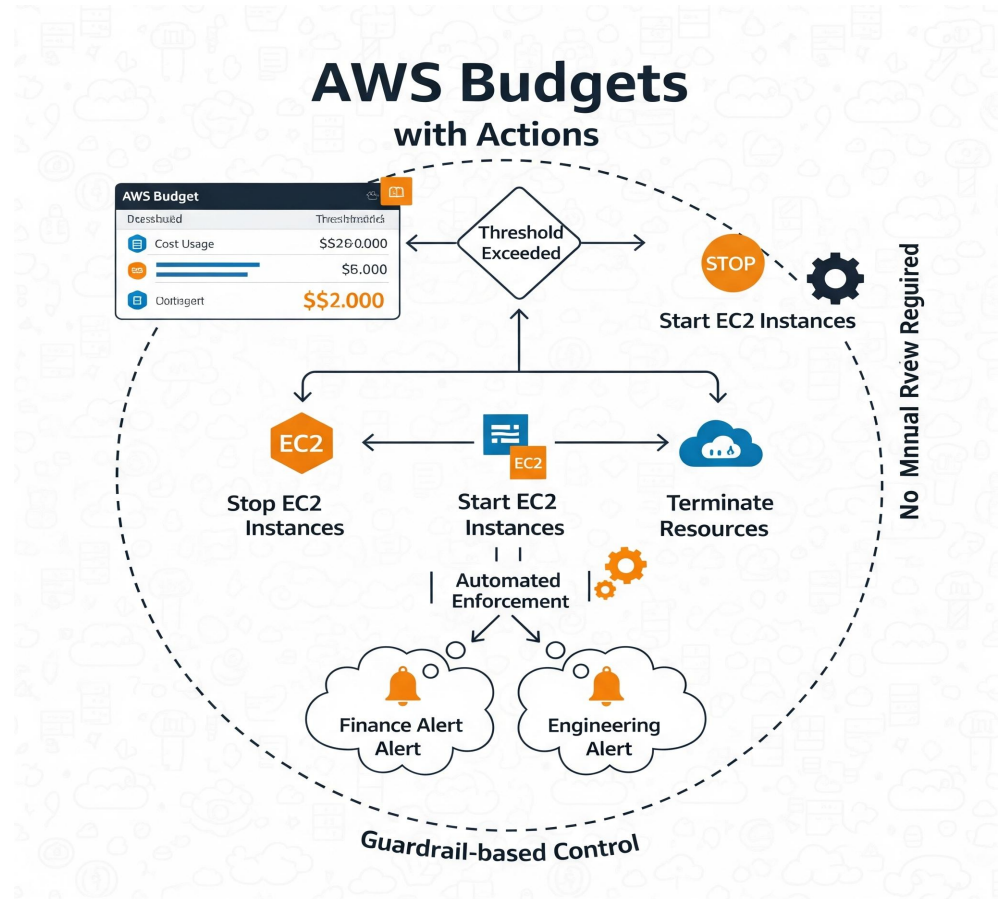
Tag Environment

- ❖ Use AWS Config + Service Catalog
- ❖ Prevent untagged resource creation
- ❖ Tags required for cost allocation
- ❖ Ensures consistent reporting
- ❖ Drives accountability



Budgets as a Governance

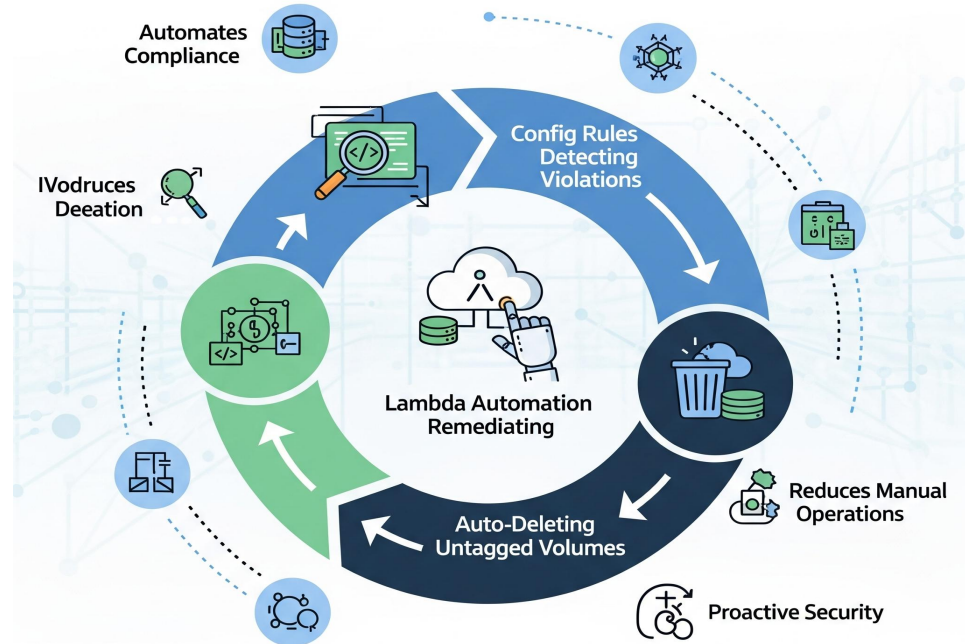
- ❖ Use AWS Budgets with actions
- ❖ Stop/start/terminate resources at thresholds
- ❖ Tie alerts to Finance & Engineering
- ❖ Enforce limits without manual review
- ❖ Builds guardrail-based control



Policy Automation

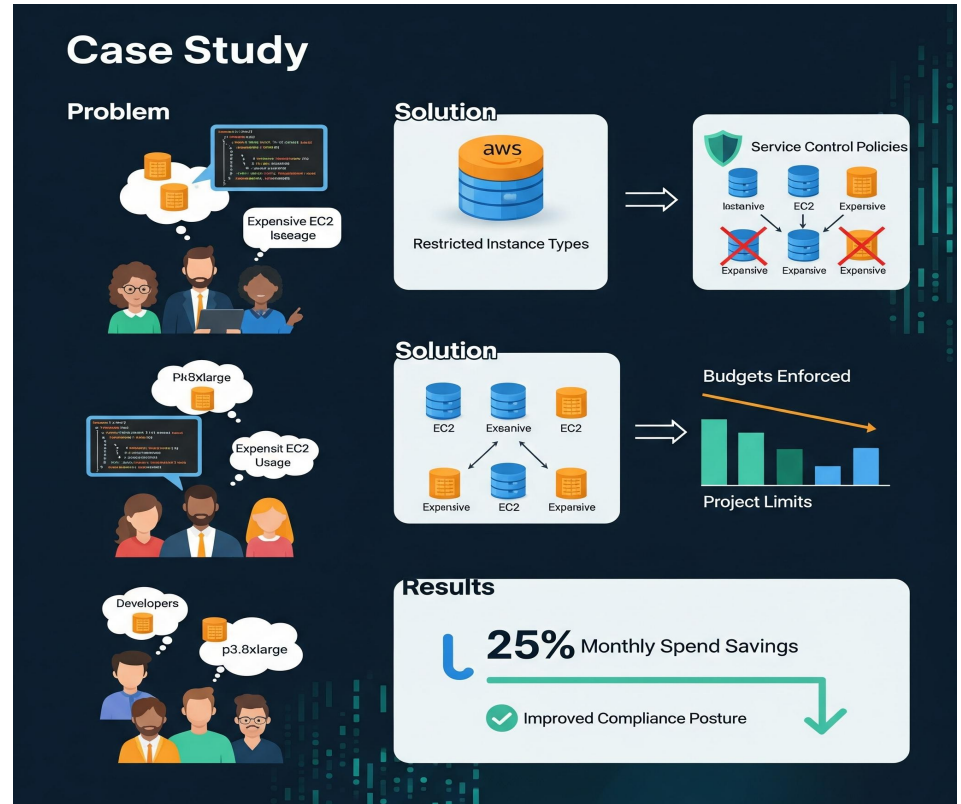
- ❖ Config rules → detect violations
- ❖ Lambda automation → remediate
- ❖ Example: auto-delete untagged volumes
- ❖ Automates compliance
- ❖ Reduces manual ops

Policy Automation



Case Study: Financial Services

- ❖ Problem: devs launched high-cost EC2 types
- ❖ SCP restricted instance types
- ❖ Budgets enforced project limits
- ❖ Saved 25% monthly spend
- ❖ Improved compliance posture



Best Practices-Governance

- ❖ Use SCP + IAM + Budgets together
- ❖ Enforce tagging consistently
- ❖ Automate remediation
- ❖ Monitor with Config & CloudWatch
- ❖ Review governance quarterly



Summary-Governance

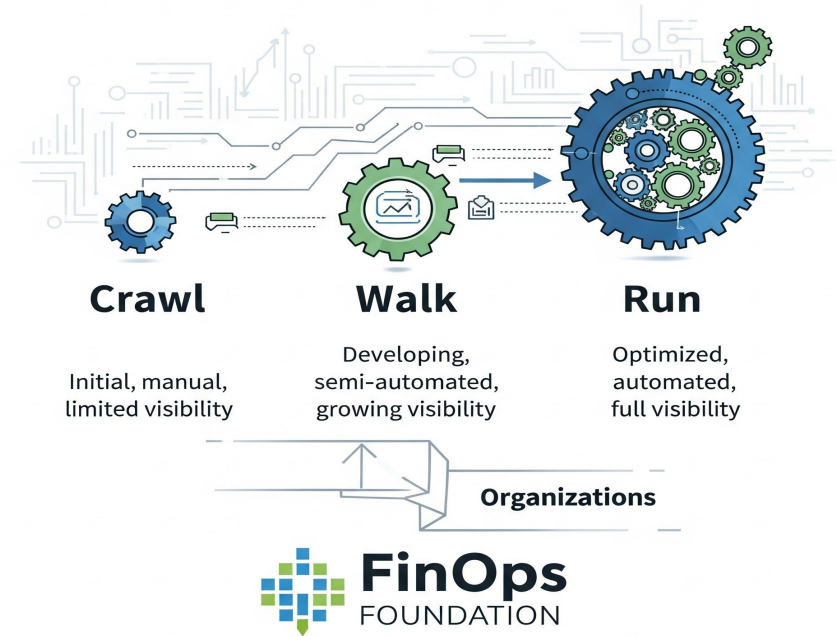
- ❖ Governance ensures accountability
- ❖ AWS provides multiple enforcement tools
- ❖ Must balance agility + control
- ❖ Automation reduces overhead
- ❖ Critical for large org FinOps

FinOps Maturity Model

Maturity Model Overview

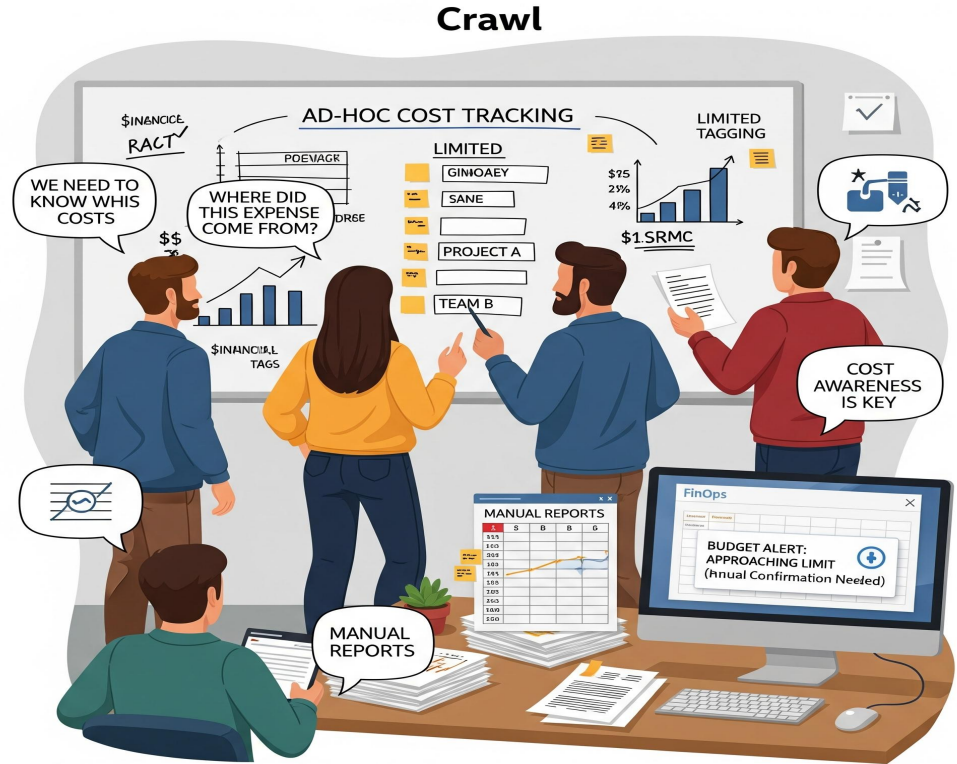
- ❖ Crawl → Walk → Run stages
- ❖ Defines FinOps adoption journey
- ❖ Assesses cultural + tooling maturity
- ❖ Guides roadmap for orgs
- ❖ Supported by FinOps Foundation

FinOps Maturity Model



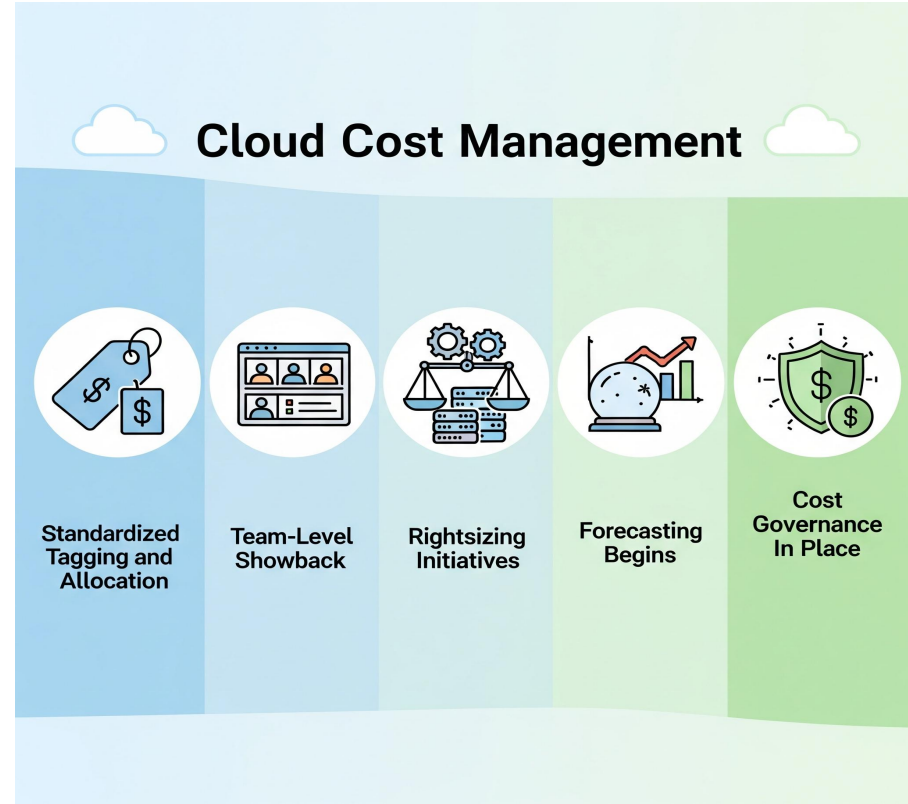
Crawl Stage

- ❖ Ad-hoc cost tracking
- ❖ Limited tagging
- ❖ Manual reports
- ❖ Budget alerts not automated
- ❖ Focus on awareness



Walk Stage

- ❖ Standardized tagging & allocation
- ❖ Team-level showback
- ❖ Rightsizing initiatives
- ❖ Forecasting begins
- ❖ Cost governance in place



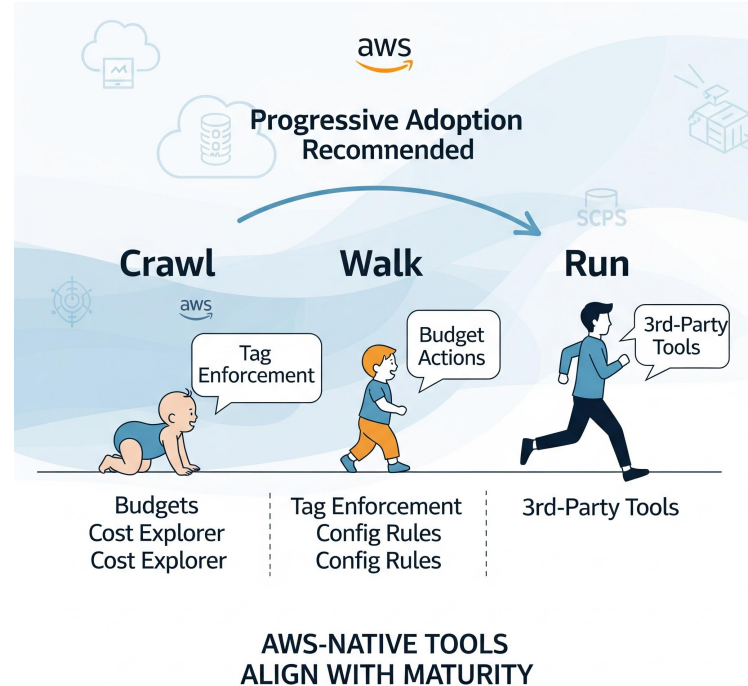
Run Stage

- ❖ Automated cost controls
- ❖ Predictive forecasting
- ❖ Business-driven KPIs
- ❖ FinOps embedded in culture
- ❖ Continuous optimization



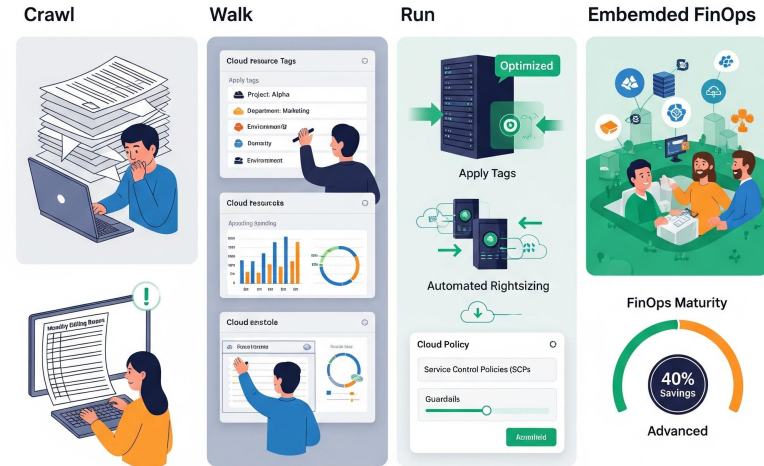
AWS Role in Crawl-Walk-Run

- ❖ Crawl: Budgets + Cost Explorer
- ❖ Walk: Tag enforcement + Config Rules
- ❖ Run: Budgets Actions + SCPs + 3rd-party tools
- ❖ AWS-native tools align with maturity
- ❖ Progressive adoption recommended



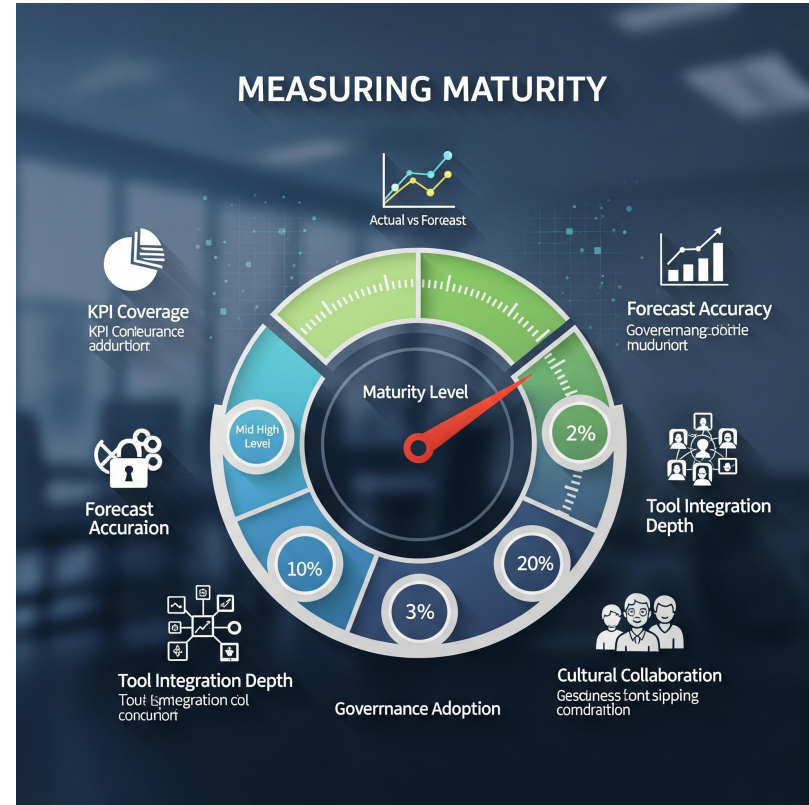
Case Study -Enterprise Crawl->Run

- ❖ Crawl: monthly billing reports
- ❖ Walk: tagging & dashboards
- ❖ Run: automated rightsizing + SCPs
- ❖ Result: 40% cost savings
- ❖ FinOps maturity embedded



Measuring Maturity

- ❖ KPI coverage
- ❖ Forecast accuracy
- ❖ Governance adoption
- ❖ Tool integration depth
- ❖ Cultural collaboration



Summary – Maturity Model

- ❖ Crawl → Walk → Run = FinOps journey
- ❖ AWS tools map to each stage
- ❖ Measured by KPIs + governance
- ❖ Case studies validate value
- ❖ Supports continuous improvement

Tools & Integrations

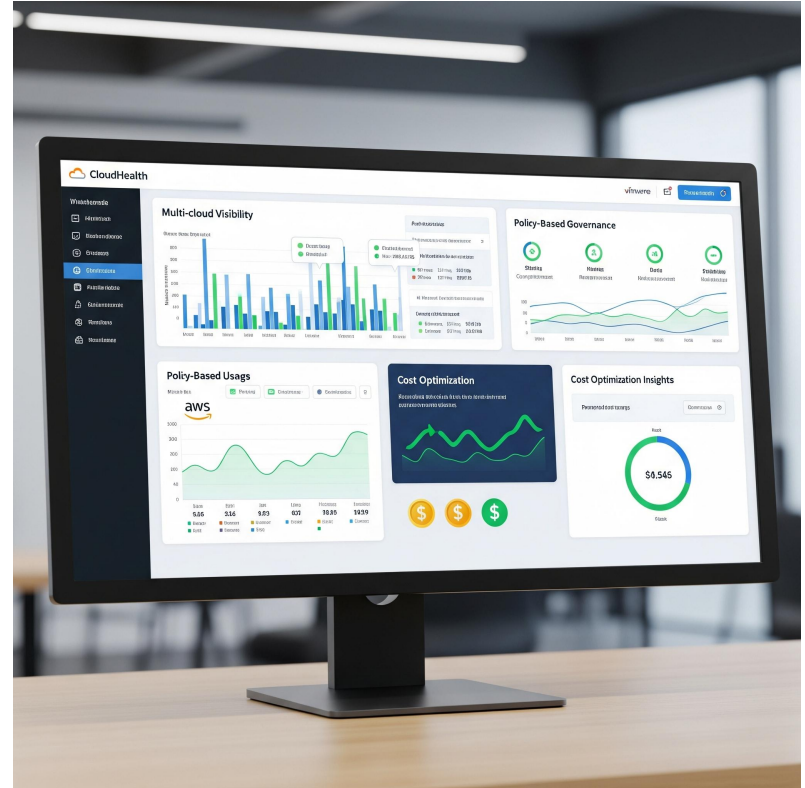
Why Tools Matter

- ❖ Enhance visibility
- ❖ Provide multi-cloud view
- ❖ Automate recommendations
- ❖ Support governance policies
- ❖ Empower cross-team collaboration



CloudHealth Overview

- ❖ VMware-owned FinOps tool
- ❖ Multi-cloud visibility
- ❖ Policy-based governance
- ❖ Cost optimization insights
- ❖ Integrates with AWS



Cloudability Overview

- ❖ Acquired by Apptio
- ❖ Detailed cost allocation
- ❖ Forecasting & budgeting features
- ❖ Unit economics tracking
- ❖ AWS-native integration

Cloudability



DETAILED COST
ALLOCATION

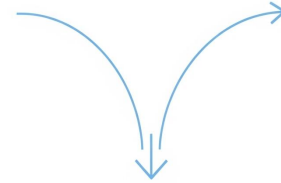
FORECASTING
& BUDGETING
FEATURES



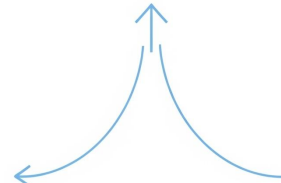
ECONOMICS
TRACKING



UNIT ECONOMICS
TRACKING



ACQUISITION BY
 **Apptio**



Overview

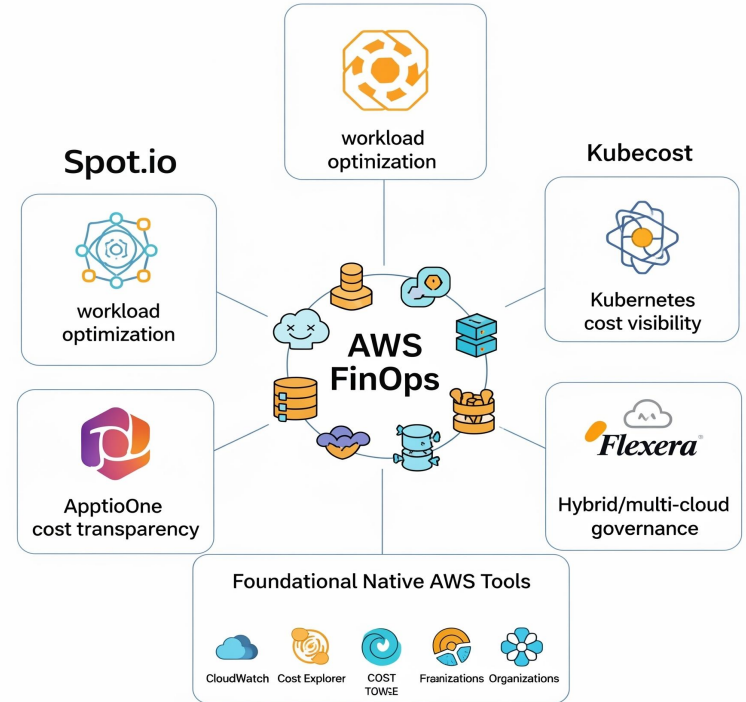


aws

AWS-NATIVE
INTEGRATION

Other Popular Tools

- ❖ Spot.io → workload optimization
- ❖ Kubecost → Kubernetes cost visibility
- ❖ ApptioOne → enterprise cost transparency
- ❖ Flexera → hybrid/multi-cloud governance
- ❖ Native AWS tools still foundational

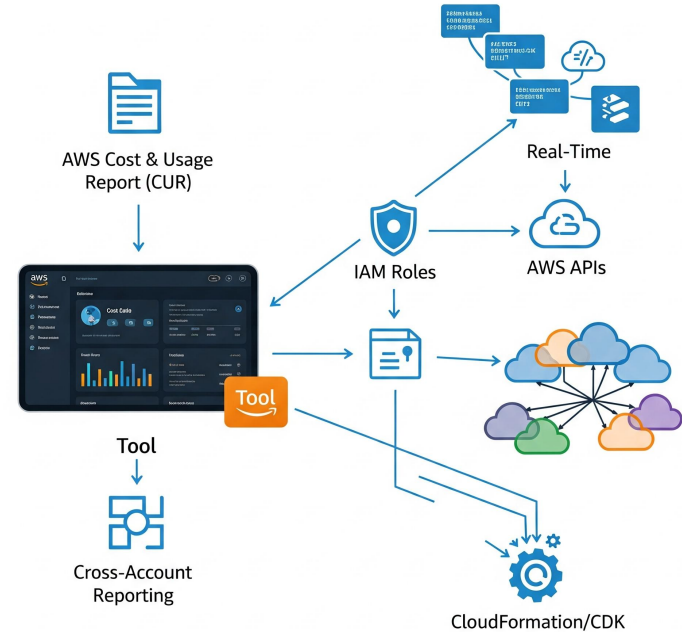


AWS-Native vs Third-Party

- ❖ AWS Budgets, Cost Explorer, Config = native
- ❖ 3rd-party adds advanced analytics
- ❖ Tradeoff: native = cheaper, simpler
- ❖ 3rd-party = cross-cloud, deeper insights
- ❖ Often used together

Tool Integration with AWS

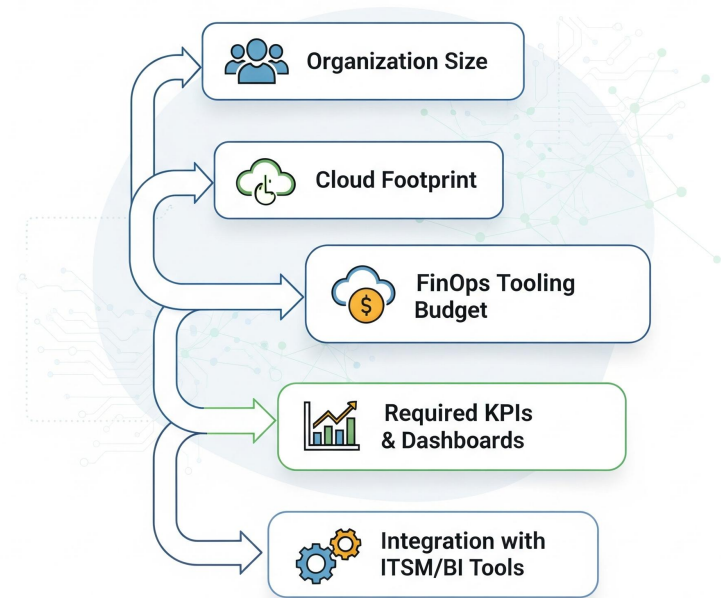
- ❖ Connect via AWS CUR (Cost & Usage Report)
- ❖ Use APIs for real-time data
- ❖ IAM roles for secure access
- ❖ Cross-account reporting supported
- ❖ Automate with CloudFormation/CDK



Tool Selection Criteria

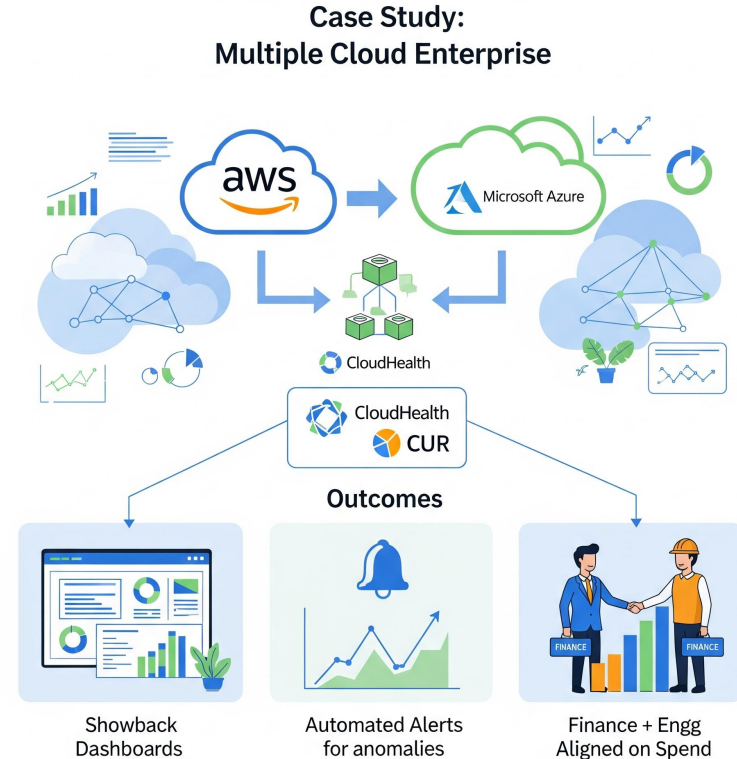
- ❖ Org size and cloud footprint
- ❖ Multi-cloud vs AWS-only
- ❖ Budget for FinOps tooling
- ❖ Required KPIs & dashboards
- ❖ Integration with ITSM/BI tools

Tool Selection Criteria



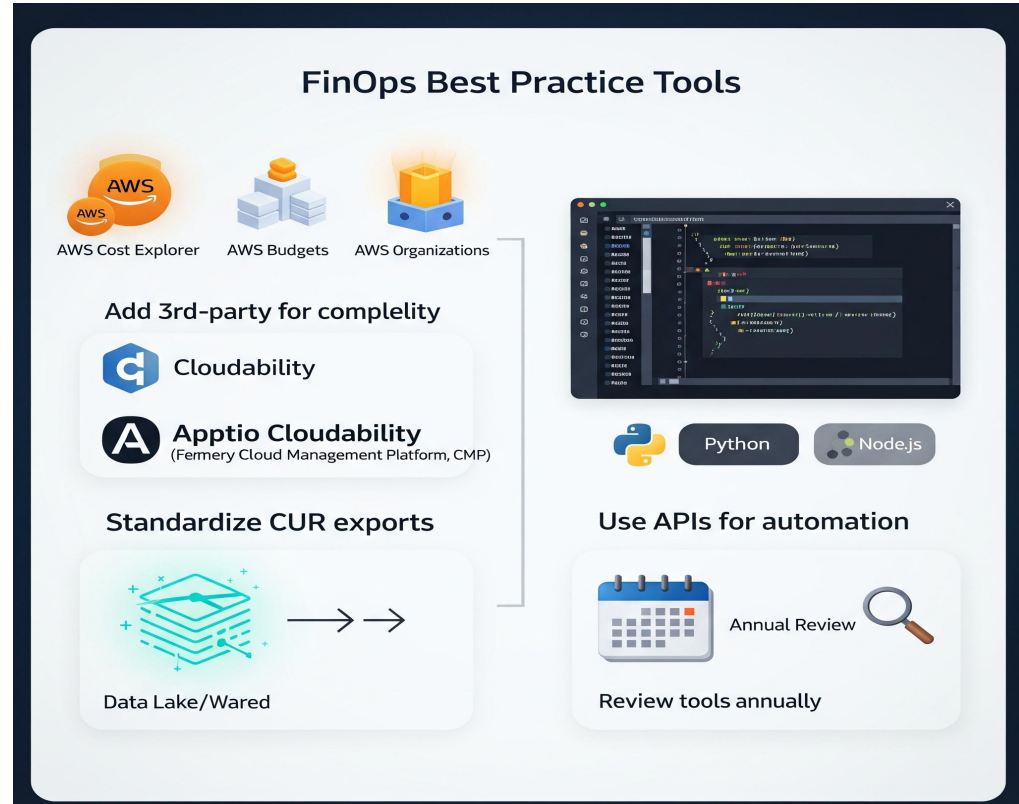
Case Study-Multi-Cloud Enterprise

- ❖ Needed AWS + Azure visibility
- ❖ Adopted CloudHealth + CUR
- ❖ Implemented showback dashboards
- ❖ Automated alerts for anomalies
- ❖ Finance + Engg aligned on spend



Best Practices-Tools

- ❖ Start with AWS-native
- ❖ Add 3rd-party for complexity
- ❖ Standardize CUR exports
- ❖ Use APIs for automation
- ❖ Review tools annually



Summary-Tools

- ❖ Tools enhance AWS FinOps maturity
- ❖ CloudHealth & Cloudability popular
- ❖ CUR is central for integration
- ❖ Selection depends on org needs
- ❖ Balance cost, insights, complexity